

Project Proposal

● Graded

Student

Josh Joseph

Total Points

1 / 1 pts

Question 1

Project Proposal

■ 1 / 1 pt

✓ + 0.5 pts Clear, detailed description of what tools/architectures will be used.

✓ + 0.5 pts Clear, detailed description of timeline on how student plans to build their idea out.

+ 0 pts Missing details on tools and plan

💬 Awesome! Think about ways you can turn it into an idea that when people hear it for the first time, they go "wow!",

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Track: Automation / Agent Systems

Project Title

NetSim: An AI-Powered Airline Network Simulator

Proposed Approach

Airlines spend millions on enterprise platforms like Sabre and Amadeus for network planning, yet their planners still fall back to spreadsheets. These tools use traditional mathematical optimization wrapped in modern UIs, and while every incumbent is investing heavily in agentic AI, all of it is pointed at customer-facing booking chatbots, not the planning workflow itself. No tool today lets a network planner ask "What happens if we add a hub in Nashville?" or "What if Southwest enters our top 3 routes?" in natural language and get an integrated answer.

NetSim brings LLM-based multi-agent reasoning to airline network planning for the first time. The system has four components:

1. Data layer - Real US DOT data: BTS DB1B origin-destination demand (40% sample of all US airline tickets) and T-100 segment capacity data, providing actual market dynamics rather than synthetic inputs.
2. Simulation engine - Models an airline network as a graph, tracking route economics (demand, load factors, revenue, operating costs), fleet assignment, and competitive dynamics.
3. Multi-agent AI layer - Four specialized agents coordinated by an orchestrator:
 - Network Planner: Identifies underserved markets and recommends routes using demand signals and competitive landscape
 - Revenue Optimizer: Evaluates route profitability given demand elasticity, aircraft type, and frequency
 - Disruption Simulator: Models weather events, aircraft groundings, or competitor entry and their cascading effects
 - Fleet Assigner: Matches aircraft types to routes based on demand, range, and operating cost
4. Natural language interface - A map-based UI where users run what-if scenarios conversationally, with agents responding in real time with analysis and visualizations. This is the core differentiator: integrated reasoning across fleet, schedule, demand, and revenue in a single interaction — something no existing tool offers.

Implementation Plan & Timeline

- Weeks 1+2: (Data pipeline) ingest and model BTS DB1B and T-100 datasets, build airport/route graph, implement route economics (cost estimation, revenue modeling, load factor calculation).
- Weeks 3+4: (Simulation engine) network graph operations, fleet type definitions, competitive dynamics modeling, scenario state management.
- Weeks 5+6: (Map-based visualization) interactive route map, network statistics dashboard, UI for scenario exploration.
- Weeks 7+8: (Multi-agent layer) implement four specialized agents with LLM-based reasoning over simulation state, natural language what-if queries, orchestration for cross-agent coordination.
- Weeks 9+10: (Integration) end-to-end scenario testing against known airline network decisions, disruption simulation, demo preparation.

